EXPERIMENT NO: 05

Experiment Title: Apply array methods and object handling; create a cart total calculator with discount logic.

Software/Tools Required:

- Any text editor / VS Code
- Web Browser (Chrome/Edge) with Developer Console

Theory

1. **JavaScript Arrays:** Arrays are high-level, list-like objects used to store multiple values in a single variable. They facilitate the efficient management, traversal, and dynamic manipulation of data collections using built-in prototype methods.

2. **Array Methods & Functional Iteration:** JavaScript provides high-order functions such as map(), filter(), reduce(), and forEach() to operate on array elements declaratively. These functional programming utilities simplify complex tasks like transforming arrays, filtering elements based on logical predicates, and accumulating values.

3. **JavaScript Objects:** Objects store data in key-value pairs and represent real-world entities. Arrays in JavaScript are specialized objects with numerical keys (indices) and a dynamic auto-updating length property.

Experiment Program Code

Code 1: Push_Pop.html

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Movie Playlist</title>
</head>
<body>
    <h2>Movie Playlist</h2>
    <script>
        let movies = ["Avengers", "Inception"];
        document.write("Original Movies : "+ movies + "<br>");

        movies.push("Titanic");
        document.write("After adding movie : "+ movies + "<br>");

        movies.pop();
        document.write("After removing movies : "+ movies+ "<br>");

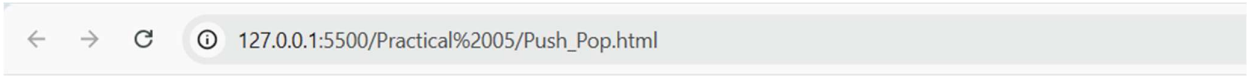
        document.write("Total movies : "+movies.length);
    </script>
</body>
</html>
```

Code 2: Reduce.html

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Document</title>
</head>
<body>
    <h2>Using Reduce Function</h2>
    <script>
        let marks = [75, 80, 65, 90, 85];
        let total = marks.reduce(function(sum, mark){
            return sum+mark;
        }, 0);

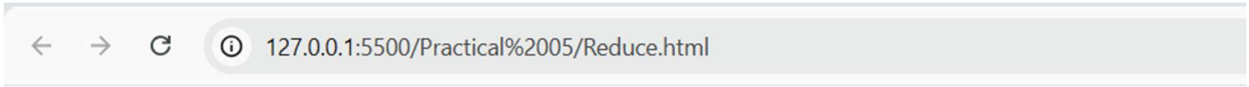
        document.write("Marks : "+ marks+ "<br>");
        document.write("Total marks: "+ total);
    </script>
</body>
</html>
```

4. Output



Movie Playlist

Original Movies : Avengers,Inception
After adding movie : Avengers,Inception,Titanic
After removing movies : Avengers,Inception
Total movies : 2



Using Reduce Function

Marks : 75,80,65,90,85
Total marks: 395

5. Case Study Title

Take an array of numbers, process string conversion to numeric elements, find the maximum and minimum values using Math.max() and Math.min() with the spread operator, and display the structured results using JavaScript object parameters.

6. Case Study Program Code

Casestudy_02_Shweta_Kushwaha_1001.html

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Array Max and Min Finder</title>
</head>
<body>

    <h2>Array Max and Min Finder</h2>

    <p>Enter comma-separated numbers (e.g., 25, 4, 98, 12, 67):</p>
    <input type="text" id="numInput" placeholder="Enter numbers here">
    <button onclick="processArray()">Find Max & Min</button>

    <div id="result"></div>

    <script>
        function processArray() {
            let rawInput = document.getElementById("numInput").value.trim();
            let resultDiv = document.getElementById("result");

            try {
                if (rawInput === "") {
                    throw new Error("Input field cannot be empty.");
                }

                // Convert comma-separated string to an array of numbers
                let stringArray = rawInput.split(",");
                let numArray = stringArray.map(item => Number(item.trim()));

                // Check for invalid numbers in array
                let hasNaN = numArray.some(num => isNaN(num));
                if (hasNaN) {
                    throw new Error("Please enter valid numeric values separated by commas.");
                }

                // Finding Max and Min values using Math methods & spread operator
                let maxVal = Math.max(...numArray);
                let minVal = Math.min(...numArray);

                // Storing operations summary inside an Object
                let report = {
                    originalArray: numArray,
                    totalElements: numArray.length,
                    maximum: maxVal,
                    minimum: minVal
                };

                // Displaying output
                resultDiv.innerHTML = `
                    <h3>Results:</h3>
                    <p><b>Array Elements:</b> [ ${report.originalArray.join(", ")} ]</p>
                    <p><b>Total Numbers:</b> ${report.totalElements}</p>
                    <p><b>Maximum Value:</b> <span style="color: green; font-weight:
bold;">${report.maximum}</span></p>
                    <p><b>Minimum Value:</b> <span style="color: red; font-weight:
bold;">${report.minimum}</span></p>
                `;

            } catch (error) {
                alert("Error: " + error.message);
                resultDiv.innerHTML = "";
```

```
            }
        }
    </script>

</body>
</html>
```

7. Output of the case study :

Name : Shweta Kushwaha PRN : 24070521001 File path :
http://127.0.0.1:5500/Practical%2002/Casestudy_05_Shweta_Kushwaha_1001.html



8. Conclusion

Thus, JavaScript array methods (split, map, some, push, shift) and object handling operations were successfully applied. A case study program was developed to take an array of numbers, process string conversion to numeric elements, find the maximum and minimum values using Math.max() and Math.min() with the spread operator, and display the structured results using JavaScript object parameters.